

Fan “Frank” XUE

PhD MSc BEng

MACM SMCGS SMIEEE MASC LMISDE MCSGPC

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Part I: Basic Information

Mission

My mission is to make buildings and cities machine-readable, machine-reasonable, and machine-verifiable, with humanity in the loop. My focus areas include:

- AEC domain knowledge × Spatial information
- Distributed nature of AEC × Distributed nurture with automation
- Student success × Interdisciplinary teamwork

Academic Qualification

- 2000 – 2004 **BEng in Automation**, College of Traffic Engineering, Civil Aviation University of China (CAUC).
- 2004 – 2007 **MSc in Computer Technology**, College of Computer Science & Technology, CAUC.
- 2007 – 2012 **PhD in Systems Engineering**, Dept of Industrial & Systems Engineering, HK PolyU.

Academic Appointments

- 2010 – 2016 **Research Assistant, Research Associate, & Postdoctoral Fellow**, HK PolyU, Hong Kong SAR.
- 2016 – 2019 **Research Assistant Professor**, Dept of Real Estate & Construction, HKU, Hong Kong SAR.
- 2019 – 2024 **Assistant Professor**, Dept of Real Estate & Construction, HKU, Hong Kong SAR.
- 2020 – 2021 **Assistant Professor (part-time)**, Dept of Urban Planning & Design, HKU, Hong Kong SAR.
- 2026 – 2026 **Academic Visiting Fellow**, Dept of Engineering, University of Cambridge, Cambridge, UK.
- 2024 –Present **Associate Professor**, Dept of Real Estate & Construction, HKU, Hong Kong SAR.
- 2025 –Present **Associate Head (Research)**, Dept of Real Estate & Construction, HKU, Hong Kong SAR.

Professional Memberships

- 2016–Present **Senior Member ('25-); Member ('16-25)**, Institute of Electrical and Electronics Engineers (IEEE).
- 2017–Present **Member**, Association for Computing Machinery (ACM).
- 2020–Present **Senior Member**, China Graphics Society (CGS).
- 2020–2024 **Member**, Hong Kong Geographic Information System Association (HKGISA).
- 2023–Present **Member**, Architectural Society of China (ASC).
- 2024–Present **Lifetime Member**, International Society for Digital Earth (ISDE).
- 2025–Present **Member**, Chinese Society for Geodesy, Photogrammetry, and Cartography (CSGPC).

Research Interests and Keywords

- Building and city information modeling (BIM/CIM) and digital twin
- 3D/4D point clouds and LiDAR
- Derivative-free and combinatorial optimization
- Construction automation and distributed systems
- Machine learning and urban informatics

Part II: Honors and Awards

External Honors, Prizes, & Awards

- 2026 **Winner, CIB Sebestyén Future Leaders Award**, *International Council for Research and Innovation in Building and Construction (CIB)*; project: *Breaking Language Barriers in Knowledge Exchange: Expert-in-the-loop Deep Translation of CIB Video for Construction 5.0*, (Awardee position: 1/4), 🏆<https://cibworld.org/cib-sebestyen-future-leaders-award-2026-winner/>.
- 2026 **Supervisor of Young Innovator (Student)**, *2025 CIC Construction Innovation Award, Construction Industry Council (CIC), Hong Kong SAR, China*, (as supervisor), 🏆<https://frankxue.com/pdf/zhou25cic.pdf>.
- 2025 **Top Cited Article**, *Journal of Industrial Ecology*, *IF = 5.7, Springer Nature*, (4/4), 🏆https://frankxue.com/pdf/xue25feature_cert.pdf.
- 2025 **Excellent Corresponding Expert [fractional Associate Editor]**, *Frontiers of Engineering Management, journal IF: 9.8, HEP & Springer*, 🏆<https://frankxue.com/pdf/xue25fem.jpg>.
- 2024 **Gold Medal**, *49th International Exhibition of Inventions of Geneva, Geneva, Switzerland*. "Automatic AI generation of photorealistic window views and openness indexes using 3D City Information Models (CIM)" (US 63/269,891, PCT/CN2023/077947, WO 2023/179296 A1), (3/3), 🏆<https://frankxue.com/pdf/li24geneva.pdf> 🏆<https://www.hku.hk/press/press-releases/detail/27236.html>.
- 2024 **Dual Winners (2D track and 3D track)**, *The 4th International Scan-to-BIM Challenge, CVPR 2024, Seattle, USA*, (4/4), (4/4), 🏆https://frankxue.com/pdf/meng24scan2bim_1st.pdf 🏆https://frankxue.com/pdf/wu24scan2cad_1st.pdf.
- 2024 **First Prize**, *The 2024 Geospatial Information Technology Progress Award (GITPA), China Association for Geospatial Industry and Sciences (CAGIS), Kunming, China*, (5/20), 🏆https://frankxue.com/pdf/xue24gis_award.pdf.
- 2024 **Dual Outstanding Mentor Award**, *The 16th Innovation Competition in Construction Engineering and Management, Tsinghua University, Beijing, and Gammon Construction Ltd., Hong Kong*, (1/1), as supervisor, 🏆<https://frankxue.com/pdf/xue24iccm.pdf> 🏆<https://frankxue.com/pdf/chen24iccm.pdf>.
- 2024 **Merit Award of Student Research (as mentor)**, *The Hong Kong OpenBIM / OpenGIS Awards 2024, Hong Kong SAR, China*, (5/5), 🏆<https://frankxue.com/pdf/kong24minimizing.pdf>.
- 2023 **Winner (3D/BIM track) and First Runner-up (2D/CAD track)**, *The 3rd International Scan-to-BIM Challenge (Workshop on CV4AEC, CVPR 2023), Vancouver, Canada*, (4/4), (4/4), Media coverage: HKU News [reposted by EurikAlert! (US), AlphaGalileo (EU) and SDGsUniversities (UN)], Sky Post (HK), Oriental Daily (HK) 🏆https://frankxue.com/pdf/wu23scan2bim_1st.jpg 🏆https://frankxue.com/pdf/wu23scan2cad_2nd.jpg.
- 2023 **First Prize**, *The 2023 Survey and Mapping Science and Technology Award, Chinese Society for Geodesy, Photogrammetry and Cartography (CSGPC), Deqing, Zhejiang, China*, (7/15), Media coverage: HKU News 🏆<https://frankxue.com/pdf/xue23csgpc.pdf> 🏆https://www.hku.hk/press/news_detail_26825.html.
- 2023 **Best Conference Paper Award (1st place)**, *Global Smart Cities Summit cum The 3rd International Conference on Urban Informatics (GSCS & ICUI 2023), Hong Kong SAR, China*, (3/3), 🏆https://frankxue.com/pdf/li23cimvw_award.jpg.
- 2023 **Outstanding Paper Award**, *The 28th International Symposium on Advancement of Construction Management and Real Estate (CRIOCM 2023), Nanjing, China*, (4/4), 🏆https://frankxue.com/pdf/meng23decoding_award.jpg.
- 2023 **Honorable Mention**, *BIM Automation Arena 2023, Hong Kong Institute of Building Information Modelling (HKIBIM), Hong Kong SAR, China*, (2/5), 🏆https://frankxue.com/pdf/chen23bim_award.pdf.
- 2023 – Now **Stanford World's Top 2% Scientist (Scopus) and Clarivate ESI Top 1% Scholar (Web of Science)**, *Stanford University / Elsevier and Clarivate Analytics*.
- 2022 **Second Place (3D track) and Third Place (2D track)**, *The 2nd International Scan-to-BIM Challenge, CVPR 2022, New Orleans, USA*, (2/2), (3/3), 🏆https://frankxue.com/pdf/wu22scan2bim_2nd.pdf 🏆https://frankxue.com/pdf/wu22scan2bim_3rd.pdf.

- 2022 **2022 Best Peer-Reviewed Paper Award**, *Journal of Management in Engineering*, IF = 6.0, American Society of Civil Engineers (ASCE), (6/6), 🏆 https://frankxue.com/pdf/wu22jme_award.pdf.
- 2022 **Merit Award of Research**, *The Hong Kong OpenBIM / OpenGIS Awards 2022*, Hong Kong SAR, China, (3/3), 🏆 <https://frankxue.com/pdf/wu22openbim.jpg>.
- 2022 **2021 Best Paper Award**, *Resources, Conservation and Recycling*, IF = 10.4, Elsevier, (3/3), 🏆 https://frankxue.com/pdf/lu21rcr_award.pdf.
- 2021 **Winner (only one worldwide)**, *buildingSMART International Awards for Professional Research 2021*, bSI Virtual Summit 2021, London, UK, (2/6), 🏆 https://frankxue.com/pdf/lu21obbi_award.jpg.
- 2021 **Second Runner-up**, *The 1st International Scan-to-BIM Challenge*, Workshop on CV4AEC, CVPR 2021, (3/3), 🏆 https://cv4aec.github.io/index_2021#challenge.
- 2021 **Outstanding Alumnus**, *Civil Aviation University of China*, (1/1).
- 2021 **Outstanding Paper Award and Dual Merit Paper Awards**, *The 26th International Symposium on Advancement of Construction Management and Real Estate (CRIOCM 2021)*, Beijing, China.
- 2020 **Early Career Scheme (ECS) Award 2019/20**, Research Grants Council (RGC) of Hong Kong SAR, (1/1), HK\$ 50,000.
- 2020 **2020 Featured Article**, *ISPRS Journal of Photogrammetry and Remote Sensing*, IF = 12.9, Elsevier, (1/4), 🏆 <https://www.journals.elsevier.com/isprs-journal-of-photogrammetry-and-remote-sensing/featured-articles/2020-featured-articles>.
- 2020 **Dual Outstanding Paper Awards**, *The 25th International Symposium on Advancement of Construction Management and Real Estate (CRIOCM 2020)*, Wuhan, China, (), (), 🏆 <https://frankxue.com/pdf/li20award.pdf> 🏆 <https://frankxue.com/pdf/laos20award.pdf>.
- 2019 **Best Paper Award**, *The 2nd International Conference on Sustainable Buildings and Structures (ICSBS 2019)*, Suzhou, China, (1/4), 🏆 <https://frankxue.com/pdf/xue19cim-albedo-award.pdf>.
- 2017 **Special Award for Joint Computing in Construction**, *The LC3 Innovation Competition 2017*, Heraklion, Greece, (3/3).
- 2017 **Finalist**, *Construction Industry Council (CIC) Construction Innovation Award 2017*, Hong Kong SAR, China, (5/10), 🏆 <https://www.youtube.com/watch?v=Ad19pDIn81I>.
- 2016 **Best Paper Award**, *The 21st International Conference on Advancement of Construction Management and Real Estate (CRIOCM 2016)*, Hong Kong SAR, China, (1/5), 🏆 <https://frankxue.com/pdf/xue16dfo-award.pdf>.
- 2016 **Dual Winners**, *The 2nd Combinatorial Black-Box Optimization Competition (CBBOC 2016)*, Denver, Colorado, USA, (1/2), (1/2).
- 2015 **Triple Winners**, *The Combinatorial Black-Box Optimization Competition (CBBOC 2015)*, Genetic and Evolutionary Computation Conference (GECCO 2015), Madrid, Spain, (1/2), (1/2), (1/2).
- 2011 **Third Runner-up (of 30 teams)**, *The 1st Cross-domain Heuristic Search Challenge (CHeSC 2011)*, The 53rd Operational Research Society Conference (OR53), Nottingham, UK, (1/3).

HKU Internal awards

- 2026 **Faculty KE Award**, *Faculty of Architecture, HKU*, (1/4), HK\$ 100,000..
- 2023 **(Co-author of) HKU Foundation Publication Award for Research Postgraduate Students 2023**, *HKU*, (2/4).
- 2022 **Merit Award in "Data is the New Art Form" Art Competition 2022**, *HKU Musketeers Foundation Institute of Data Science*, (1/1).
- 2020 **Research Output Prize 2018/19**, *Faculty of Architecture, HKU*, (1/4), HK\$ 120,000..

PhD Students' Awards (as supervisor)

🎓 Mr Maosu (Eric) Li (2020-24), later Assistant Professor at HKUST (Guangzhou)

- 2024 **Gold Medal**, *The 49th International Exhibition of Inventions of Geneva*, (Team awardee), CIM window-view invention 🏆 <https://frankxue.com/pdf/li24geneva.pdf>.

- 2024 **Excellent Dissertation Award**, *IEEE GRSS Hong Kong Chapter*, (1/1), 🏆https://frankxue.com/li24thesis_award.jpg.
- 2024 **HKU Foundation Award for Outstanding Research Postgraduate Students 2023/24**, *The University of Hong Kong*, (1/1), PhD.
- 2024 **Best Poster Presenter**, *HKU Research Postgraduate Symposium 2024*, (1/1), PhD, Yr-4 🏆https://gradsch.hku.hk/news_and_events/news_and_future_events/symposium-2024/best-poster-presenter-awards.
- 2024 **Talent Development Scholarship**, *Education Bureau, Government of the Hong Kong SAR*, (1/1).
- 2024 **URC Postdoctoral Fellowship**, *The University of Hong Kong*, (1/1), 2024–2025.
- 2023 **Best Conference Paper Award (1st place)**, *GSCS & ICUI 2023, Hong Kong*, (First author), 🏆https://frankxue.com/pdf/li23cimvw_award.jpg.
- 2023 **First Prize, 2022 Smart City Research and Innovation Scheme**, *International Society for Urban Informatics (ISUI)*, (1/3), 🏆<https://frankxue.com/pdf/li23isui.pdf>.
- 2023 **HKU Foundation Publication Award for Research Postgraduate Students**, *HKU Foundation*, (First author and recipient), ≤20 awardees/year 🏆https://frankxue.com/pdf/li23pub_award.jpg.
- 2023 **Research Postgraduate Student Innovation Award 2022/23 (HK\$100,000)**, *The University of Hong Kong*, (1/1), 10 awardees/year 🏆https://gradsch.hku.hk/news_and_events/news_and_future_events.
- 2022 **Second Place (3D) and Third Place (2D)**, *2nd International Scan-to-BIM Challenge, CVPR 2022, New Orleans*, (Team member), with Y. Wu 🏆<https://cv4aec.github.io/>.
- 2022 **Asia-Pacific Economic Cooperation (APEC) Scholarship 2021/22**, *Government of the Hong Kong SAR*, (1/1).
- 2022 **Talent Development Scholarship (HK\$10,000)**, *Education Bureau, Government of the Hong Kong SAR*, (1/1), PhD, Yr-3.
- 2021 **Champion, 2021 Esri Young Scholars Award**, *Esri China (Hong Kong)*, (1/1), story map on 3D accessibility to nature 🏆<https://bit.ly/32yz8xL>.
- 🎓Ms Yijie Wu, 2021–25, later Postdoc Fellow at HKPolyU
- 2025 **Li Ka Shing Prize 2024–25 (HK\$20,000)**, *The University of Hong Kong*, (1/1), Top 1% PhD honor at HKU; A prestigious award, 6 awardees/year from ~850 graduates 🏆https://frankxue.com/pdf/wu26lks_award.png.
- 2023 **Winner (3D track) and Second Place (2D track)**, *3rd International Scan-to-BIM Challenge, CVPR 2023*, (Team lead), methods SBASE-MO / FLKPP++ 🏆<https://cv4aec.github.io/>.
- 2023 **HKU Foundation Publication Award for Research Postgraduate Students**, *HKU Foundation*, (Co-author), ≤20 awardees/year 🏆https://frankxue.com/pdf/li23pub_award.jpg.
- 2023 **Talent Development Scholarship (HK\$10,000)**, *Education Bureau, Government of the Hong Kong SAR*, (Awardee), PhD, Yr-2.
- 2022 **Second Place (3D) and Third Place (2D)**, *2nd International Scan-to-BIM Challenge, CVPR 2022, New Orleans*, (Team lead), method FLKPP 🏆<https://cv4aec.github.io/>.
- 2022 **Merit Award of Research**, *The Hong Kong OpenBIM / OpenGIS Awards 2022*, (Awardee), 🏆<https://frankxue.com/pdf/wu22openbim2022.pdf>.
- 2021 **Second Runner-up, 1st International Scan-to-BIM Challenge**, *CVPR 2021*, (Team lead), 🏆<https://cv4aec.github.io/#challenge>.
- 🎓Mr Rui (Jeremy) Zhao, 2021–25, later Assistant Professor at Macao UST
- 2024 **Merit Award of Student Research**, *The Hong Kong OpenBIM / OpenGIS Awards 2024*, (Team member), 🏆<https://frankxue.com/pdf/kong24minimizing.pdf>.
- 2024 **Participation Award**, *16th ICCEM, Tsinghua University and Gammon Construction Ltd.*, (Team member), 🏆<https://frankxue.com/pdf/zhao24iccem.jpg>.
- 2022 **Best Peer-Reviewed Paper Award**, *Journal of Management in Engineering, ASCE*, (Co-author), 🏆https://frankxue.com/pdf/wu22jme_award.pdf.

2021 **Winner, Professional Research**, *buildingSMART International Awards 2021, London*, (Team member), OBBI project 🏆<https://www.buildingsmart.org/award-winners-2021/>.

🎓Mr Leo (Leo) Liang, 2021-25, later Postdoc Fellow at HKPolyU

2025 **Talent Development Scholarship (HK\$10,000)**, *Education Bureau, Government of the Hong Kong SAR*, (1/1).

2024 **First Prize, 16th ICCEM**, *Tsinghua University and Gammon Construction Ltd.*, (Team member), 🏆<https://frankxue.com/pdf/chen24iccem.pdf>.

2023 **Winner (3D track)**, *3rd International Scan-to-BIM Challenge, CVPR 2023*, (Team member), 🏆<https://cv4aec.github.io/>.

2022 **HKU Foundation Postgraduate Fellowship (UPF)**, *The University of Hong Kong*, (1/1), 🏆https://scholar.aas.hku.hk/?action=showonesscheme&ss_id=421.

🎓Mr Zhe (Philo) Chen, 2022-25, later Postdoc Fellow at HKU

2025 **Talent Development Scholarship (HK\$10,000)**, *Education Bureau, Government of the Hong Kong SAR*, (Awardee).

2024 **First Prize, 16th ICCEM**, *Tsinghua University and Gammon Construction Ltd.*, (Team member), ranked 2/72 🏆<https://frankxue.com/pdf/chen24iccem.pdf>.

2023 **Honorable Mention, BIM Automation Arena 2023, HKIBIM**, (Awardee), 🏆<https://www.hkibim.org/index.php?mid=53>.

Mr Lingming (Karle) Kong, 2022-26

2026 **Young Innovator (Student)**, *2025 CIC Construction Innovation Award (HK\$100,000)*, (2/2), with Q. Zhou; Bridging LLM Agent and OpenBIM; 🏆<https://frankxue.com/pdf/zhou25cic.pdf>.

2024 **First Prize, 16th ICCEM**, *Tsinghua University and Gammon Construction Ltd.*, (Team member), ranked 2/72 🏆<https://frankxue.com/pdf/chen24iccem.pdf>.

2024 **Merit Award of Student Research**, *The Hong Kong OpenBIM / OpenGIS Awards 2024*, (Awardee), 🏆<https://frankxue.com/pdf/kong24minimizing.pdf>.

2023 **Honorable Mention, BIM Automation Arena 2023, HKIBIM**, (Awardee), 🏆<https://www.hkibim.org/index.php?mid=53>.

2022 **Gold Award, Smart Mobility**, *Hong Kong ICT Awards 2022, OGCIO*, (Team awardee), 🏆https://www.hkictawards.hk/award_en.php?year=2022.

2022 **Merit Award**, *The Hong Kong OpenBIM / OpenGIS Awards 2022*, (Team member), 🏆<https://frankxue.com/pdf/wu22openbim2022.pdf>.

Ms Jiajia Wang, 2022-26

2024 **Winner (3D track)**, *4th International Scan-to-BIM Challenge, CVPR 2024*, (Team member), 🏆<https://cv4aec.github.io/>.

2024 **Merit Award of Student Research**, *The Hong Kong OpenBIM / OpenGIS Awards 2024*, (Team member), 🏆<https://frankxue.com/pdf/kong24minimizing.pdf>.

2024 **Participation Award**, *16th ICCEM, Tsinghua University and Gammon Construction Ltd.*, (Team member).

Ms Qianyun (Joanna) Zhou, 2023-27

2026 **Young Innovator (Student)**, *2025 CIC Construction Innovation Award (HK\$100,000)*, (1/2), with L. Kong; Bridging LLM Agent and OpenBIM; 🏆<https://frankxue.com/pdf/zhou25cic.pdf>.

2023 **Second Place (2D track)**, *3rd International Scan-to-BIM Challenge, CVPR 2023*, (Team member), 🏆<https://cv4aec.github.io/>.

2023 **Honorable Mention, BIM Automation Arena 2023, HKIBIM**, (Awardee), 🏆<https://www.hkibim.org/index.php?mid=53>.

Mr Sou-Han (Hank) Chen, 2023-27

2024 **Winner (3D track)**, *4th International Scan-to-BIM Challenge, CVPR 2024, Seattle*, (Team member), 🏆https://frankxue.com/pdf/meng24scan2bim_1st.pdf.

2023 **Second Place (2D track)**, *3rd International Scan-to-BIM Challenge, CVPR 2023*, (Team member), 🏆<https://cv4aec.github.io/>.

- 2023 **Dual Honorable Mentions**, *BIM Automation Arena 2023, HKIBIM*, (Awardee), 🏆 <https://www.hkibim.org/index.php?mid=53>.

Mr Meng (Simon) Sun, 2023-27

- 2025 **Talent Development Scholarship 2025/26**, *Education Bureau, Government of the Hong Kong SAR*, (Awardee).
- 2025 **Best Individual Performance Award**, *The 4th Financial Career Accelerator*, (Awardee).
- 2024 **Winner (2D track)**, *4th International Scan-to-BIM Challenge, CVPR 2024*, (Team member), 🏆 <https://cv4aec.github.io/>.
- 2024 **Participation Award**, *16th ICCM, Tsinghua University and Gammon Construction Ltd.*, (Team member).

Ms Siyuan Meng, 2024-28

- 2026 **First Runner-up**, *2026 Esri Young Scholars Award, Esri*, (Awardee), 🏆 <https://storymaps.arcgis.com/stories/9a434eae210f458ea30217046699ea6c>.
- 2025 **Talent Development Scholarship 2025/26**, *Education Bureau, Government of the Hong Kong SAR*, (Awardee).
- 2024 **Winner (3D track)**, *4th International Scan-to-BIM Challenge, CVPR 2024, Seattle*, (Team member), 🏆 https://frankxue.com/pdf/meng24scan2bim_1st.pdf.
- 2023 **Winner (3D track)**, *3rd International Scan-to-BIM Challenge, CVPR 2023*, (Team member), 🏆 <https://cv4aec.github.io/>.
- 2023 **Outstanding Paper Award**, *CRIOCM 2023, Nanjing*, (First author), decorative-pattern extraction for heritage DTs 🏆 https://frankxue.com/pdf/meng23decoding_award.jpg.

Mr Longyong Wu, 2025-29

- 2024 **Winner (2D track)**, *4th International Scan-to-BIM Challenge, CVPR 2024*, (Team member), 🏆 https://frankxue.com/pdf/meng24scan2bim_1st.pdf.

Bachelor & Master Theses Awards (as supervisor)

🎓 Master of Science in Construction Project Management

- 2023 **Merit Award, Dissertation/Thesis Awards for Postgraduate Students**, *The Hong Kong Institute of Surveyors (HKIS)*, (1/1), Ms Ngai, Nam Hei (Nancy)'s MSc thesis; later study at Harvard GSD 🏆 <https://www.linkedin.com/feed/update/urn:li:activity:7211574735657168896/>.

🎓 Bachelor of Science in Surveying

- 2026 **HKIS Scholarship**, *The Hong Kong Institute of Surveyors*, (1/1), Chan, Chun Hei (Henry).
- 2025 **Merit Award of QSD, Dissertation Awards**, *The Hong Kong Institute of Surveyors (HKIS)*, (1/1), Kwok Wai Shing (Jacky) 🏆 <https://frankxue.com/thumbnail/kwok26hkis.jpg>.
- 2025 **HKIS Scholarship**, *The Hong Kong Institute of Surveyors*, (1/1), Lee, Wai Hung (Sammy).
- 2024 **Top Prize (Quantity Surveying), Outstanding Dissertation Awards**, *The Hong Kong Institute of Surveyors (HKIS)*, (1/1), Ng, Kai Yui (Clive).
- 2024 **Finalist (Planning and Development), Outstanding Dissertation Awards**, *The Hong Kong Institute of Surveyors (HKIS)*, (1/1), Mok, Kai Ho (Gordon).
- 2022 **Merit Prize, OSH Best Project Scholarship 2021–22**, *Occupational Safety and Health Council (OSHC)*, (1/1), Lee, Hok Kiu (Chris).
- 2021 **Second Award (General Practice), Outstanding Final Year Dissertation Awards**, *The Hong Kong Institute of Surveyors (HKIS)*, (1/1), Li, Yuet Tat (Jeff).

Research Team



The research team of postdocs, postgraduates, RAs led by Frank (Fall 2024)

Part III: Research and Scholarship

Publications

Indexing	110+ papers indexed by Web of Science (SCI/SSCI/ISTP) Core Collection	Source: Web of Science
Quality	About 90% of papers published in world Q1 (top 25%) journals	Source: Scopus
Highly-cited	Clarivate's ESI Top 1% Scholar; 6 ESI Highly Cited Papers; Citations 9,800	Source: ESI
	Stanford World's Top 2% Scientists;	Source: Elsevier
<i>h</i> -index	52 Source: Google Scholar, 46 Source: Scopus, or 41 Source: Web of Science	
Fields Impacts	Field-Weighted Citation Impact (FWCI) = 3.05 (305% of all subjects); by Subject: 3.5 / Engineering, 3.8 / Computer, 3.3 / Environment.	Source: SciVal

Books

- 2026 Crolla K, **Xue F**, et al. (eds.). *Computer-Aided Architectural Design. Catalytic Interfaces*. Cham, Switzerland: Springer. ISBN: 9783032396846. in press.
- 2025 Crolla K, **Xue F**, et al. (eds.). *CAAD Futures 2025 Catalytic Interfaces: Conference Proceedings (Volume I)*. The University of Hong Kong. ISBN 9789887834939. doi:10.25442/hku.29265815
- 2025 Crolla K, **Xue F**, et al. (eds.). *CAAD Futures 2025 Catalytic Interfaces: Conference Proceedings (Volume II)*. The University of Hong Kong. ISBN 9789887834946. doi:10.25442/hku.29364860
- 2024 Li Z, **Xue F**, et al. (eds.). *Construction Robotics* (in Chinese). China Construction Industry Publishing. ISBN 9787112310500
- 2023 Li Z, Ding C, **Xue F**, et al. (eds.). *Prefabricated Lean Construction* (in Chinese). China Construction Industry Publishing. ISBN 9787112286591.

Book chapters

- 2022 **Xue F**. As-built building information modeling: joint effort of 3D reconstruction and semantic enrichment. In *Research Companion to Building Information Modeling*, Chap 10, pp. 190–211. Edward Elgar Publishing. doi:10.4337/9781839105524.00019

Journal special issues

- 2025 Wang Z, Grieves M, Wen Y, **Xue F**. *Special Issue on Digital Twins*. IEEE Internet Computing. doi:10.1109/MIC.2025.3540532

Journal papers

- 2026 You K, Chen K & **Xue F***. Automated scan-to-BIM for construction digital transformation: Conceptual framework, processing methods and best-practice guidelines. *Automation in Construction*, 182, 106755. doi:10.1016/j.autcon.2025.106755
- 2026 Zhao R, Kong L, Sun M & **Xue F***. A Blockchain 4.0-Digital Twin paradigm for prediction and reasoning in distributed construction project management. *Computers & Industrial Engineering*, 220, 112253. doi:10.1016/j.cie.2026.112253
- 2026 Meng S, Wu L, Li M, Yeh AGO & **Xue F***. Automated window detection for digital twin buildings: A generalized 'sandwich' model and practical guidelines. *Energy and Buildings*, 360, 117384. doi:10.1016/j.enbuild.2026.117384
- 2026 Liang D, Wu L, Sun M, Hu R, Kong L, Pan Y & **Xue F***. Transfer learning from building information model-based synthetic data for three-dimensional module detection in point clouds of modular-integrated construction hoisting. *Engineering Applications of Artificial Intelligence*, 164, 113243. doi:10.1016/j.engappai.2025.113243
- 2026 Wei X, **Xue F***, Wang J & Liang D. Bi-objective optimization of low-carbon, low-cost scaffolding design using bamboo and steel with a study in China. *Resources, Conservation & Recycling Advances*, 30, 200334. doi:10.1016/j.rcradv.2026.200334
- 2026 Sun M, Liang D, Wang J, Bate B & **Xue F***. Clarity in DEM cementation predictions: Integrating automated machine learning and interpretability analysis of shear wave velocity. *Advanced Engineering Informatics*, 69, 104060. doi:10.1016/j.aei.2025.104060

- 2026 Meng S, Li M* , Xie J & **Xue F**. Multi-perspective assessment of greenery visibility from streets, windows, and drones using deep learning and CIM. *International Journal of Geographical Information Science*, in press. (🏆First Runner-up of 2026 ESRI Young Scholars Award) doi:10.1080/13658816.2026.2667807
- 2026 Li M, **Xue F** & Yeh AGO*. Citywide assessment of perceived window view openness using photorealistic 3D model imagery and automatic machine learning. *Building and Environment*, 305, 115234. doi:10.1016/j.buildenv.2026.115234
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- 2022 Kong L, Chen C, Zhao R, Chen Z, Wu L, Yang Z, Li X, Lu W & **Xue F**. When permissioned blockchain meets IoT oracles: An on-chain quality assurance system for off-shore modular construction manufacture. In *2022 IEEE 1st Global Emerging Technology Blockchain Forum (iGETblockchain)*. doi:10.1109/iGETblockchain56591.2022.10087164
- 2022 Wu Y, Li M & **Xue F**. Floor layer-based kernels and pillars of points (FLKPP). In *2nd Workshop on Computer Vision in the Built Environment, CVPR 2022, New Orleans, USA*. 🏆 https://frankxue.com/pdf/wu22scan2bim_2nd.pdf First runner-up (3D) / second runner-up (2D), 2nd International Scan-to-BIM Challenge
- 2021 Li CZ, Chen Z, **Xue F**, Xiao B & Tam VWY. Cloud asset-enabled integrated blockchain platform for prefabricated construction management. In Guo H et al. (eds) *Proceedings of the 26th International Symposium on Advancement of Construction Management and Real Estate (CRIOCM 2021)*, Springer LNOR. Outstanding Paper Award; doi:10.1007/978-981-19-5255-5
- 2021 Zhao R, Chen Z, Wu Y, Lou J, Wu L & **Xue F**. Comparison of BIM collaboration paradigms for digital twin readiness: Centralized files, decentralized clouds, and distributed blockchains. In *Proceedings of the 38th International Conference of CIB W78*, Luxembourg, 315–325. doi:10.46421/2706-6568.38.2020.paper032
- 2021 Wu Y & **Xue F**. FloorPP-Net: Reconstructing floor plans using point pillars for Scan-to-BIM. In *1st International Scan-to-BIM Challenge, Workshop on CV4AEC, CVPR 2021*. Second runner-up; arXiv:2106.10635
- 2021 Lou J, Xu J, Lu W & **Xue F**. A generic framework for BIM component naming. In Guo H et al. (eds) *Proceedings of the 26th International Symposium on Advancement of Construction Management and Real Estate (CRIOCM 2021)*, Springer LNOR. doi:10.1007/978-981-19-5255-5
- 2020 Lee WMW, Lu W & **Xue F**. Understanding construction waste recycling in Hong Kong: SWOT analysis of the government's prevailing initiatives. In Lu X et al. (eds) *Proceedings of the 25th International Symposium on Advancement of Construction Management and Real Estate (CRIOCM 2020)*, Springer. 🏆 <https://frankxue.com/pdf/lee20award.pdf> Merit Paper Award; doi:10.1007/978-981-16-3587-8
- 2020 Digital twinning construction objects: Lessons learned from pose estimation methods. In *Joint Conference ICCCB E and CIB W78 2020*.
- 2020 Indoor-outdoor navigation without beacons: Compensating smartphone AR positioning errors with 3D pedestrian networks. In *Joint Conference ICCCB E and CIB W78 2020*.
- 2019 **Xue F**, Chen K, & Lu W. Semantic enrichment of city information models with LiDAR-based rooftop albedo, In *Proceedings of the 2nd International Conference on Sustainable Buildings and Structures (ICSBS 2019)*, (🏆 <https://frankxue.com/pdf/xue19cim-albedo-award.pdf> Best Paper Award), Ch. 27, CRC Press. doi:10.1201/9781003000716-27
- 2019 **Xue F**, Chen K, & Lu W. Understanding unstructured 3D point clouds for creating digital twin city: An unsupervised hierarchical clustering approach, In *Proceedings of CIB World Building Congress 2019*. doi:e-Proceedings
- 2018 **Xue F**, Chiaradia AJF, Webster CJ, . . . , & Lu W. Personalized walkability assessment for pedestrian paths: An as-built BIM approach using ubiquitous augmented reality (AR) smartphone and deep transfer learning, In *Proceedings of the 23rd International Symposium on the Advancement of Construction Management and Real Estate*. Pre-print
- 2018 **Xue F**, Chen K, & Lu W. Architectural symmetry detection from 3D urban point clouds: A derivative-free optimization (DFO) approach. In *Proceedings of the 35th CIB W78 2018 Conference: IT in Design, Construction, and Management*, pp. 513–519, Springer, Cham. doi:10.1007/978-3-030-00220-6_61
- 2017 Chen K, **Xue F**, & Lu W. Development of 3D building models using multi-source data: A Study of high-density urban area in Hong Kong. In *Lean and Computing in Construction Congress - Joint Conference on Computing in Construction (JC3 2017)*, 609–616, doi:10.24928/JC3-2017/0252

- 2017 **Xue F**, & Shen G. Design of an efficient hyper-heuristic algorithm CMA-VNS for combinatorial black-box optimization problems. In *Proceedings of GECCO Companion*, 1157–1162, doi:10.1145/3067695.3082054
- 2016 **Xue F**, Chen, K, Liu, D, ..., & Lu W. An optimization-based semantic building model generation method with a pilot case of a demolished construction. In *Proceedings of the 21st International Conference on Advancement of Construction Management & Real Estate*, Chap. 22, 231–241. Springer Singapore. ( <https://frankxue.com/pdf/xue16dfo-award.pdf> Best Paper Award) doi:10.1007/978-981-10-6190-5_22

Patents

- 2025 **Xue F** et al. System and method for BIM family model retrieval based on building's 3D point cloud features. CN 202510171724.0.  *filing*
- 2024 **Xue F** et al. Real-time modular construction hoisting monitoring method and device based on 4D lidar point cloud and image fusion. CN 202411492723.8.  *filing*
- 2024 **Xue F** et al. 4D point cloud and tension information-based monitoring device and method for arch rib rotation construction. CN 202411590045.9.  *filing*
- 2024 **Xue F** et al. Optimization System and Method for multiple Web GPU service configuration based on large foundation model. CN 202411777369.3.  *filing*
- 2024 Yeh AGO, Li M & **Xue F**. System and methods for quantifying and calculating window view openness indexes. Family:  US 63/269,891; PCT/CN2023/077947; WO 2023/179296 A1; CN 202380029205.X →  CN 119234216 A. *filing*
- 2023 **Xue F**. E-inspection system for offsite construction manufacturing and transport. US Patent 12,236,313 B2.  *Awarded*
- 2023 **Xue F** et al. Building component quality control system and method based on digital twin. CN 202311704690.4.  *filing*
- 2023 **Xue F** et al. Web System and Method for Structural Maintenance and Monitoring of Modular-Integrated Construction. CN 202311611591.1.  *filing*
- 2022 **Xue F**, Yeh AGO, Wu Y & Chen Z. Indoor 3D barrier-free mapping based on LiDAR point cloud and BIM clash simulation. CN 202210303655.0; PCT/CN2022/084953.  *Awarded*
- 2022 **Xue F**, Yeh AGO, Chen Z & Zhao R. A cultural relics traceability system for museums. CN 202210303663.5; PCT/CN2022/084959.  *filing*
- 2022 **Xue F**, Yeh AGO, Wu Y & Yang Z. An analytics system for museum touring based on BIM and surveillance videos. CN 202210302636.6; PCT/CN2022/084962.  *Awarded*
- 2021 **Xue F** & Chen Z. A 3D quality inspection equipment for prefabricated construction products. CN 202120404023.4.(utility model)  *Awarded*
- 2021 **Xue F** & Chen Z. A quality assurance management information system for prefabricated construction products. CN 202120417423.9.  *filed; failed*

Other publications: Open source scientific software libraries

- 2024 Open BIM exchange on Blockchain 3.0 virtual disk. <https://github.com/KarleKong/openBIMdisk>, in Python & JavaScript
- 2024 CIM-WV: A 2D semantic segmentation dataset of rich window view contents in high-rise, high-density Hong Kong based on photorealistic City Information Models. <https://doi.org/10.25442/hku.24647487>
- 2023 APA-like citation template with DOI for MS Word bibliography. <https://github.com/ffxue/word-bibliography-apalike-doi>, in XSLT language
- 2023 HRHD-HK: A Benchmark Dataset of High-Rise and High-Density Urban Scenes for 3D Semantic Segmentation of Photogrammetric Point Clouds. <https://doi.org/10.25442/hku.23701866>, world only 3D dataset for high-rise high-density cities
- 2022 ChainPM: a Blockchain 3.0 paradigm for digital twins in construction project management. <https://github.com/jeremyRZ/ChainPM>, in Python language
- 2020 RegARD: Registration based on Architectural Reflection Detection. <https://github.com/iiiiijiii/RegARD>, in Python language

- 2019 Content Analysis by Latent Semantic Analysis. <https://github.com/ffxue/ca-lsa>, in Python language
- 2018 ODAS: Optimization-based Detection of Architectural Systems. <https://github.com/ffxue/odas>, in C++ language
- 2018 COBIMG: Constrained Optimization-based Building Information Model Generator. <https://github.com/ffxue/cobimg>, in C++ and Ruby language
- 2017 CMA-VNS/CMA-VNS2: An efficient hyper-heuristic algorithm for combinatorial black-box optimization problems. <https://github.com/ffxue/cmavns>, in C++ language

Plenary/Invited Lectures in International/Regional Conferences

- 2026 **From Static to Dynamic: Temporal-Spectral-Structural Point Clouds for Digital Twin Construction**, *Department of Engineering, University of Cambridge, Cambridge, UK.*
- 2025 **Automated Scan-to-BIM for Precise 3D Modeling of Built Assets**, *RILS Public Lecture, HKPolyU, Hong Kong.*
- 2025 **Application of Ground-Penetrating Radar in Concrete Heritage Buildings: A Pilot**, *2025 Symposium for Spatial Information-enabled Cultural Heritage Conservation, Macau University of Science and Technology, Macao.*
- 2024 **Digital Heritage Conservation With Spatial Information Technologies**, *Xianghai Forum, HKUST, Hong Kong.*
- 2024 **Smart Construction Course Development and Teaching at HKU**, *2nd National Forum on Programme Development of Smart Construction, SCUT, Guangzhou, China.*
- 2023 **Fully Automatic Scan-to-BIM: Consolidation of unsupervised, supervised, and reinforced learning**, *The 2023 Smart City and Smart Construction Forum cum 2023 Smart Construction Conference of Architectural Society of China, Wuhan, China.*
- 2023 **Bridging the gap between point clouds for GeoAI: Role of supervised, reinforced and unsupervised learning**, *Workshop of GeoAI and Big Data for Urban, Environment, and Sustainability, HKPolyU, Hong Kong.*
- 2023 **Design for eXcellence (DfX) with Digital Twins: From Reality Data to Semantic Models to Optimized Design**, *Faculty of Architecture and the Built Environment, TU Delft, Netherlands.*
- 2021 **CIM-enabled quantitative view assessment in architectural design and space planning**, *The 38th International Symposium on Automation and Robotics in Construction (ISARC2021), Dubai, UAE, (As the top-quality paper by anonymous peer-review).*
- 2020 **Blockchain for Construction in the Post-COVID-19 Era**, *The 6th China BIM Conference – Youth Forum, Taiyuan, China.*

Part IV: External Peer-Reviewed Competitive Research Grants (HK\$ 18M)

As Principal Investigator (PI) (My individual research, HK\$ 4.5M)

- 2026 – 2028 **HK RGC General Research Fund (GRF) (17201325)**, *Architectural openings detection from 3D point clouds: A bi-objective optimization approach addressing local hollow semantic graph and global design regularity*, HK\$ 822,264.
- 2024 – 2026 **HK RGC General Research Fund (GRF) (17200123)**, *As-built building information modeling (BIM) of RC details from radar images: A derivative-free optimization (DFO) approach based on Yeas model*, HK\$ 793,117.
- 2023 – 2025 **Guangdong Natural Science Fund (GDNSF)**, *From 3D Real Scene to 3D semantics: Reconstruction of semantic volumetric building models using 3D skeletons in urban point clouds*, HK\$ 112,000.
- 2022 – 2024 **HK RGC General Research Fund (GRF) (17200221)**, *Building information modeling (BIM) on chain: from stocks of model files to flows of semantic trees*, HK\$ 799,692.
- 2021 – 2023 **HK RGC Early Career Scheme (ECS) (27200520)**, *From point cloud to building and city information model (BIM/CIM): A study of architectonic grammar optimization*, HK\$ 635,569.
- 2018 – 2020 **HK RGC General Research Fund (GRF) (17201717)**, *Generation of semantically rich as-built Building Information Models (BIMs): A derivative-free optimization approach*, HK\$ 454,157.
- 2018 – 2021 **HK RGC General Research Fund (GRF) (17200218)**, *A derivative-free optimization (DFO) approach to architectural symmetry detection from 3D point clouds*, HK\$ 522,846.

As Project Coordinator (PC) (Leader of a major research, HK\$ 6.5M)

- 2023 – 2025 **HK ITC Innovation and Technology Fund (ITF) Tier-1 (ITP/004/23LP)**, *Scan-to-BIM Automation System for Built Assets Digitalization in Hong Kong*, Accepted; HK\$ 5,210,000 from funder + HK\$ 2,300,000 industry matching.
- 2021 – 2023 **Shenzhen-Hong Kong-Macau Technology Research Programme (Type C) (SGDX20201103093600002)**, *High-definition 3D point cloud and dynamic semantics enabled lean quality assurance system for prefabricated housing products*, HK\$ 1,260,000.

As Co-PI (Leader of a work package in a major research, HK\$ 5M out of 69M)

- 2025 – 2027 **HK ITC Innovation and Technology Fund (ITF) Tier-1 (ITP/041/24LP)**, *Large Language Models and Knowledge Graphs Enabled IoT-BIM Platform for Production Scheduling in Modular Integrated Construction (MiC)*, HK\$ 1M of HK\$ 4.92, (PC: Prof Geoffrey Shen).
- 2024 – 2026 **HK ITC Innovation and Technology Fund (ITF) PSTS (ITT/004/24LP)**, *e-TranStar 2.0: i-Core-enabled Smart Just-in-Time MiC Transportation Planning*, HK\$ 0.8M of HK\$ 3.73M, (PC: Prof Wilson Lu).
- 2023 – 2025 **Collaborative Research Fund (CRF) (C7080-22GF)**, *Generative DfX in high-rise modular building: An expert-augmented cascade graph learning and optimisation approach*, HK\$ 0.8M of HK\$ 5,310,000, (PC: Prof W Lu).
- 2022 – 2026 **HK RGC Theme-based Research Scheme (TRS) (T22-504/21-R)**, *Healthy and resilient city with pervasive LoCHs*, HK\$ 1,347,000 of HK\$ 34,600,000, (PC: Prof J Niu).
- 2021 – 2023 **HK ITC Innovation and Technology Fund (ITF) Tier-1 (ITP/029/20LP)**, *"BIM Square": Blockchain and i-Core-enabled Multi-stakeholder Building Information Modelling Platform for Construction Logistics and Supply Chain Management in Hong Kong*, HK\$ 1.06M of HK\$ 10,360,714.25, (PC: Prof Wilson Lu).
- 2019 – 2022 **Key R&D Program of the Guangdong Province (2019B010151001)**, *The Applications of Virtual Reality Technologies for Cultural Heritage Conservation in the Guangdong-Hong Kong-Macao Greater Bay Area*, 5% of CNY 10,000,000, (PC: Dr Youyi Xiong).

As Co-I (Participant and consultant, HK\$ 0 out of 10M)

- 2017 – 2020 **NSFC General Program (71671156)**, *A Social Network Analysis and NK-model based study of stakeholder relationship in mega construction projects*, CNY 493,000, (PI: Prof Geoffrey Shen).
- 2018 – 2022 **NSSFC Major Program (17ZDA062)**, *Evaluation on the environment-resources carrying capacity in large Chinese cities and policy instruments with the big data background*, CNY 600,000, (PI: Prof Liyin Shen).
- 2019 – 2020 **HK Public Policy Research (PPR) (2018.A8.078.18D)**, *“Finding the Needles in a Haystack”: Identification of the Illegal Dumping of Construction Waste Using Big Data*, HK\$ 447,350, (PI: Prof Wilson Lu).
- 2019 – 2021 **HK Strategic Public Policy Research (SPPR) (S2018.A8.010.18S)**, *Boosting Construction Waste Material Sharing in the Guangdong-Hong Kong-Macao Bay Area*, HK\$ 2,902,000, (PI: Prof Wilson Lu).
- 2020 – 2022 **HK Environment & Conservation Fund (ECF) (111/2019)**, *Big data-based “AI inspector” for gauging inert contents at the off-site construction waste sorting facilities in Hong Kong*, HK\$ 485,000, (PI: Prof Wilson Lu).
- 2021 – 2024 **NSFC General Program (72071022)**, *Unfolding the multiple complexities in the formation, quantification, and interactions of resource metabolism in construction projects*, CNY 480,000, (PI: Prof Chris Hong).
- 2021 – 2023 **HK RGC General Research Fund (GRF) (17203920)**, *Enhancing Underground Development Users’ Health through an Automated Risk Assessment System for Facilities Management*, HK\$ 684,520, (PI: Prof Isabelle Chan).
- 2022 – 2023 **HK Environment & Conservation Fund (ECF) (128/2021)**, *Building demolition waste management through smart BIM in Hong Kong*, HK\$ 478,130, (PI: Prof Ray Zhong).
- 2023 – 2024 **HK Public Policy Research Funding Scheme (PPR) (2022.A8.129.22C)**, *Prefabrication Hub Feasibility for Hong Kong Construction Industry: An Integrated Approach*, HK\$ 244,949, (PI: Prof Ray Zhong).
- 2023 – 2025 **HK RGC General Research Fund (GRF) (15225822)**, *Stakeholder Analysis in the Dynamic and Complex Environment of Megaprojects*, HK\$ 1,115,452, (PI: Prof Geoffrey Qiping Shen).
- 2023 – 2025 **HK RGC General Research Fund (GRF) (15219422)**, *Inspection work package on chain: dynamic planning and accountable collaboration*, HK\$ 786,000, (PI: Prof Xiao Li).
- 2024 – 2025 **NSFC/RGC JRS (Conference) (N_HKU7130/24)**, *Frontiers of Digital Twins in Intelligent Manufacturing and Smart Cities*, HK\$ 244,000, (PI: Prof Ray Zhong).
- 2026 – 2027 **HK Environment & Conservation Fund (ECF) (253/2025)**, *AIRSORT: A lightweight AI system for composition recognition to facilitate construction waste separation at source*, HK\$ 845,738, (PI: Prof Wilson Lu).
- 2025 – 2027 **HK RGC General Research Fund (GRF) (17202224)**, *PanoCause: Auto-profiling illegal dumping ‘black spots’ using a novel panoptic deep causal discovery approach*, HK\$ 755,442, (PI: Prof Wilson Lu).
- 2025 – 2027 **HK RGC General Research Fund (GRF) (17202424)**, *Detecting and modeling building defects in context: A novel learning-based registration of as-damaged reality capture with as-designed drawings*, HK\$ 784,135, (PI: Dr Junjie Chen).
- 2025 – 2028 **NSSFC Major Program (25&ZD115)**, *Research on coordination mechanisms for the conservation and revitalization of urban cultural heritage*, CNY 800,000, (PI: Prof Anrong Dang).
- 2026 – 2028 **HK RGC General Research Fund (GRF) (17212725)**, *A novel blockchain smart contract and zero-knowledge proof approach for accurate and secure access to building information models for multi-stakeholder collaboration*, HK\$ 755,442, (PI: Prof Mike Wu).

As Scholarship Account Holder (HK\$ 2.2M)

- 2024 **Sino-British Fellowship Trust Visitorship**, *Visiting Postdoctoral Fellowship*, HK\$ 45,000, (Scholar: Dr Yuandong Pan).
- 2024 – 2025 **HK ITC Researcher Talent Hub (PiH/252/24, PiH/243/25)**, *Postdoctoral Fellowship for ITP/004/23LP*, HK\$ 429,525 + 922,500, (Scholar: Dr Ke You).

- 2024 – 2025 **HK ITC Researcher Talent Hub (InP/141/24)**, *Research Fellowship for ITP/004/23LP*, HK\$ 219,765, (Scholar: Mr Yibo Wang).
- 2026 – 2028 **Hong Kong Scholars Program (XJWQ2025008; 2025-277)**, *3D/4D Point Cloud Processing*, CNY 360,000 (CNPC, MOHRSS) +HK\$ 219,600 (URC).

HKU internal (selected from HK\$ 4M)

- 2026 **Doris Zimmern HKUCambridge Hughes Hall Fellowships**.
- 2026 **Visiting Professors Scheme 2025-26 (Prof. J.E. Stoter)**, HK\$ 51,200.
- 2021 – 2025 **HKU Research Postgraduate (RPg) Central Pool 2021/22**, *Blockchain 4.0-based building information modelling (B4BIM)*, HK\$ 530,000.
- 2020 – 2021 **HKU Small Equipment Grant (102009916.097506)**, *Mobile Color LiDAR Scanners*, 60% of HK\$ 760,000.

Part V: Teaching and Learning

Particulars of courses

As coordinator and teacher

2019 – 2026	RECO3032/39 , <i>Research methods</i> ,	BSc(Surveying).
2020 – 2026	RECO7613 , <i>Information Technology in Design and Construction</i> ,	MSc(IPD).
2022 – 2026	RECO2041 , <i>Information Tech for Urban Development</i> ,	BSc(Surveying).
2022 – 2026	RECO6022 , <i>Research Seminar II</i> ,	PhD/MPhil.
2025 – 2026	CCST6003 , <i>Digital Twinning Buildings and City with Minecraft Worlds</i> , undergraduate.	University-wide

As teacher only

2018 – 2019	RECO6004 , <i>Construction Economics</i> ,	MSc(CPM).
2020 – 2026	RECO7097 , <i>Modern Development in Construction</i> ,	MSc(CPM).
2020 – 2026	RECO4001 , <i>Dissertation</i> ,	BSc(Surveying).
2020 – 2026	RECO4002/12 , <i>Studio V</i> ,	BSc(Surveying).
2020 – 2023	RECO7607 , <i>Understanding Industry Practice</i> ,	MSc(IPD).
2020 – 2023	RECO7608 , <i>Future Industry Directions</i> ,	MSc(IPD).
2022 – 2022	CONS3006 , <i>Professional Practice in Architectural Conservation</i> ,	BA(Conservation).
2024 – 2025	CONS8224 , <i>Digital Heritage: Theory and Application</i> ,	MSc(Conservation).
2021 – 2023	URBA6402 , <i>Smart Planning and Design Studio</i> ,	MSc(Urban Analytics).
2025 – 2025	CVIL6060 , <i>Operation and Maintenance of Building and Civil Engineering Works</i> ,	MSc(Engineering).

As coordinator only

2021 – 2022	RECO3033 , <i>Professional Practice</i> ,	BSc(Surveying).
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Teaching innovation and curriculum development

As PI/coordinator

2021 – 2022	HKU Common Core Micro Credential (CCST6003) , <i>Digital Twinning Buildings and City with Minecraft Worlds (CCST6003)</i> , HK\$ 253,840.
2024 – 2025	HKU Teaching Development Grant (TDG) , <i>AR-GPR: Effectiveness of Augmented Reality-enabled "See-through" 3D Models of Ground-penetrating Radar for Teaching Non-destructive Testing of Reinforced Concrete</i> , HK\$ 300,000.
2021 – 2022	HKU Teaching Development Grant (TDG) , <i>Multi-user Internet Narrative Environment of HKU (MineHKU) for smart city courses and virtual campus events</i> , HK\$ 294,000.
2020 – 2021	Revamping of RECO7613 , <i>Information Technology in Design and Construction (6 credits)</i> , with experiential learning and teamworking, for MSc (IPD) Year-2 students.
2022 – 2023	Proposal and design of RECO2041 , <i>Introduction to Information Technology for Urban Development (6 credits)</i> , with experiential learning, for BSc (Surveying) Year-2 students.

As Co-I

2021 – 2023	UGC Special Grant for Strategic Development of Virtual Teaching and Learning , <i>Interactive Learning and Teaching of Construction Management in 3D Virtual Reality Environments</i> , HK\$500,000, (PI: Prof. Isabelle Chan).
2022 – 2023	HKU Teaching Development Grant (TDG) , <i>Aligning in-house problem-solving project experience with problem-based learning (PBL) for the post-pandemic era</i> , HK\$ 300,000, (PI: Dr Katherine Deng).
2023 – 2024	HKU Teaching Development Grant (TDG) , <i>MetaClass: Metaverse and robotics-enabled curriculum and pedagogical innovations to promote construction engineering and management knowledge teaching and learning</i> , HK\$ 230,040, (PI: Dr Junjie Chen).

Part VI: Knowledge Exchange

Knowledge exchange awards and projects

- 2026 KE Award 2026 (Faculty of Architecture, HKU): MiC Trio: A Building Information Modeling (BIM) and Blockchain-enabled Platform for Modular Integrated Construction (MiC) Digital Transformation. (2025/26). HK\$ 100,000, as Co-PI, PI: Prof Wilson Lu
- 2024 – 2025 KE Impact Project: Synergizing Experts' Visual Inspections and AR-enabled "See-through" 3D Models (AR-GPR) for Proactive Maintenance and Preservation of Aged Buildings and Historic Structures. (KE-IP-2024/25-8). HK\$ 150,000, as Co-I, PI: Dr Katherine Deng
- 2021 – 2022 KE Impact Project: Visualizing and Recollecting the Heritage Waterworks on Hong Kong Island and Kowloon Peninsula for the Public and the Professionals. (KE-IP-2021/22-10). HK\$ 150,000, as Co-I, PI: Dr Katherine Deng

Public lectures and seminars

- 2025 **Seminar**, *Invited by CIC Conference on Low-Altitude Economy in Construction*, Low-Altitude Economy Application in Construction: Frontiers and Future.
- 2024 **Seminar**, *Invited by Baosteel Construction Tech Ltd., Shanghai*, 4D Point Clouds and Industrial Digital Twin.
- 2024 **Lecture**, *Invited by 10th Summer School of Smart Construction Frontiers*, Digital Twin Construction: Leveraging BIM semantics with 3D/4D LiDAR point clouds, HUST, Wuhan, China.
- 2024 **Seminar**, *Invited by LSGI Research Seminar*, ML with Structured (Geo)Spatial Information in Processing Unstructured 3D Point Clouds, HKPolyU, Hong Kong.
- 2024 **Lecture**, *Invited by FCE Lunchtime Lecture*, Digital Construction: An optimization perspective, HKPolyU, Hong Kong.
- 2022 **Seminar**, *Invited by South China University of Technology, Guangzhou, China*, Sustainability, Intelligence, and Resilience (SIR) Frontiers: Symmetry and Similarity Detection for Urban Point Cloud Understanding.
- 2022 **Lecture**, *Invited by The Joint 8th Summer School of Frontiers in Construction Management and Cloud Summer Camp of College Students in Civil Engineering*, Scan-to-BIM: Reverse Engineering of construction from Point Clouds.
- 2022 **Lecture**, *Invited by HKABAEIMA*, Low-cost Digital Twins of Built Assets: Automatic creation of photo-realistic openBIM by integrating ubiquitous Augmented Reality and 2D CAD drawings, (Training Class 51, CPD event for local professionals).
- 2022 **Seminar**, *Invited by Leica Geosystems Hong Kong, Seminar "Are you ready for autonomous future?"*, 25 Nov 2022, Sheraton Hotel, Hong Kong SAR, Digital Twinning buildings and cities with 3D point clouds: A semantics perspective.
- 2020 **Lecture**, *Invited by The Joint 6th Summer School of Frontiers in Construction Management and Cloud Summer Camp of College Students in Civil Engineering*, Wuhan, China, Blockchain for Smart Construction.
- 2019 **Seminar**, *Invited by Huazhong University of Science and Technology (HUST), School of Civil Engineering and Mechanics*, From geometric landscape to fitness landscape: As-built BIM reconstruction through optimization.
- 2019 **Seminar**, *Invited by Civil Aviation University of China (CAUC), College of Computer Science & Technology*, Evolutionary computation with applications in 3D urban reconstruction.
- 2018 **Lecture**, *Invited by Zhejiang University of Finance & Economics*, Demystifying BIM & Innovation.
- 2018 **Lecture**, *Invited by Renmin University of China (RUC), School of Public Administration and Policy*, Research Methodology, Linking RFID to BIM: A demystification of research methods.
- 2018 **Lecture**, *Invited by Shenzhen University (SZU), China. College of Civil Engineering*, Frontiers in Construction Management, Semantic enrichment for BIM and GIS: A computational perspective.

Part VII: Community Service

Service to professional organizations

- 2025–2028 **Chair of Work Group of Hong Kong-Macao-Taiwan**, *Architectural Society of China (ASC) Smart Construction Scientific Committee*.
- 2025–207 **Chair**, *Appeal Panel of Smart Site Safety System Labelling Scheme (AP4SLS), Hong Kong Construction Industry Council (CIC)*.
- 2025–Present **Committee**, *Committee of Cultural Heritage Conservation, Chinese Society for Geodesy, Photogrammetry, and Cartography (CSGPC)*.
- 2024–Present **Committee**, *3rd Committee of Digital Heritage, International Society for Digital Earth (ISDE)*.
- 2023–2028 **Committee (the only member from HK until 2026)**, *Architectural Society of China (ASC) Smart Construction Scientific Committee*.
- 2021–2025 **Vice-Chair**, *ACM HK Chapter*.
- 2020–Present **Committee**, *China Graphics Society (CGS) BIM Chapter*.
- 2020–Present **Volunteer**, *IEEE Blockchain Community*.
- 2020–Present **Contact person (Faculty of Architecture, HKU)**, *Hong Kong Construction Industry Council (CIC) BIM Network*.
- 2017–2021 **Secretary**, *Association for Computing Machinery (ACM) Hong Kong Chapter*.

Service to the academia

- 2024–2028 **Panel member (Engineering)**, *APSF, Hong Kong Research Grants Council (RGC)*.
- 2024 **Panel member**, *Learning Program Accreditation, Hong Kong Council for Accreditation of Academic and Vocational Qualifications (HKCAAVQ)*.
- 2018–Present **Funding Referees**, *Various national research fundings, e.g., BELSPO (Belgium), CONICYT (Chile), Avanza UC (Chile)*.
- 2020–Present **PhD Referees**, *Various PhD theses examinations*.
RMIT (Australia) × 2, UAuckland (New Zealand) × 1, HKUST × 1, HKPolyU × 4, CityU × 1, BondU (Australia) × 1

Service to HKU

- 2026–Present **Chair of DRPC**, *Department of Real Estate and Construction, HKU*.
- 2025–Present **Associate Head (Research)**, *Department of Real Estate and Construction, HKU*.
- 2025–2026 **Member of Teacher Panel**, *Disciplinary Committee, HKU*.
- 2017–Present **Vice Director**, *HKUrbanLabs-iLab, FoA, HKU*.
- 2024–Present **Personal Data Protection Coordinator**, *Departmental Research Postgraduate Committee, Department of Real Estate and Construction, HKU*.
- 2024–Present **Vice Director**, *National Center of Technology Innovation for Digital Construction - Hong Kong Branch, HKU*.
- 2023–Present **Vice Director**, *Hong Kong-Zhuhai-Macao Research Station (HKU), Key Scientific Research Base of Application of Spatial Information Technologies in Cultural Heritage Conservation (Tsinghua), National Cultural Heritage Administration (NCHA), China*.
- 2021–2025 **Director of Research Postgraduate Studies**, *Department of Real Estate and Construction, HKU*.
- 2021–Present **Committee**, *Departmental Research Postgraduate Committee, Department of Real Estate and Construction, HKU*.

Service to international journals' editorial boards

- 2022–Present **ENGINEERING Management (former Frontiers of Engineering Management) (ISSN 2095-7513, Impact Factor = 7.7, Q1)**, *Special Corresponding Expert (fractional Associate Editor), Administered by Chinese Academy of Engineering (CAE), Co-published by Springer and Higher Education Press (HEP), China*.

2022–Present **International Journal of Crowd Science (ISSN 2398-7294, EI)**, *Associate Editor*, Co-published by IEEE and Tsinghua U Press.

2024–2025 **IEEE Internet Computing (ISSN 1941-0131, Impact Factor = 3.7, Q1)**, *Guest Editor of special issue on Digital Twins*, IEEE, <https://doi.org/10.1109/MIC.2025.3540532>.

Service to international journals as referees (ordered by counts)

(Note: The complete list of 50+ referee journals, 182 peer reviews, is available on [Web of Science](#).)

- 38 Automation in Construction
- 8 International Journal of Construction Management
- 8 Advanced Engineering Informatics
- 6 Engineering, Construction and Architectural Management
- 6 IEEE Transactions on Geoscience and Remote Sensing
- 6 Remote Sensing
- 5 IEEE Transactions on Automation Science and Engineering
- 5 Geo-Spatial Information Science
- 5 Building and Environment
- 4 IEEE Access
- 4 Journal of Computational Design and Engineering

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